

Vatsal Vaghasiya

AI/ML Engineer | Machine Learning Engineer | Data Scientist

Bengaluru, Karnataka, India | +91 8780335009 | kvaghasiya057@gmail.com

linkedin.com/in/vatsal-vaghasiya | github.com/Vatsal057 | vatsal057.github.io | kaggle.com/vatsalvaghasiya

PROFESSIONAL SUMMARY

Machine learning engineer and MTech Data Science student with 8 months of data science internship experience, two research papers under review, and 19 shipped projects across LLMs and RAG, computer vision, and MLOps. Builds ML systems end to end: data pipelines and feature engineering, model training, evaluation and calibration, REST API deployment with FastAPI and Docker, and production drift monitoring. 1st place at the Karnataka Education Datathon 2026 for an analysis of 1.38 million student records. Seeking AI/ML engineer or data scientist roles.

TECHNICAL SKILLS

Programming: Python, SQL, Dart, Swift, Kotlin, TypeScript, Shell

Machine Learning and Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, CNNs, Transformers, DeBERTa, CLIP, Siamese networks, NLP, computer vision, Grad-CAM, model calibration (temperature scaling, ECE), OpenCV, MediaPipe

LLMs, NLP and RAG: retrieval-augmented generation, sentence-transformers embeddings, vector search (ChromaDB), chunking and retrieval evaluation, prompt engineering, Whisper ASR, Tesseract OCR, Hugging Face

MLOps and Deployment: FastAPI, REST APIs, Docker, Docker Compose, CI/CD with GitHub Actions, model monitoring, data drift detection (PSI), Streamlit, Gradio, Hugging Face Spaces, Cloudflare Workers, Firebase, Git, SQLite, PostgreSQL

Data Analysis and Statistics: pandas, NumPy, SciPy, ETL, feature engineering, OLS regression, hypothesis testing, clustering (K-Means, DBSCAN, hierarchical), Power BI, Plotly, matplotlib, seaborn, Jupyter

Application Development: Flutter, SwiftUI, AppKit, Firebase (Firestore, Authentication), PyQt6

PROFESSIONAL EXPERIENCE

Data Science Intern - Amar Infotech, Ahmedabad, India | Jan 2025 - Jun 2025

- Built Python data pipelines preprocessing 50,000+ records, reducing data inconsistencies by 35%.
- Trained and tuned predictive ML models in TensorFlow and scikit-learn, reaching 87% accuracy and lowering false positives by 22%.
- Built 5+ interactive Power BI dashboards tracking business KPIs and live model performance for stakeholders.
- Automated data extraction workflows, cutting manual processing time by 40%.

Data Science Intern - iTechBrains, Ahmedabad, India | Jun 2024 - Jul 2024

- Analysed complex datasets using supervised and unsupervised learning techniques to support business decisions.
- Performed data extraction, cleaning and transformation with Python and SQL.
- Delivered data visualizations that communicated findings to non-technical stakeholders.

RESEARCH PUBLICATIONS

Efficient LLM Preference Classification Through Position Bias Mitigation and Architectural Symmetry - First author (with N. Kshetrimayum, B. Prabadevi, B. R. Prathap) | Under review, 2026

- Built a 71M-parameter Siamese DeBERTa-v3 model that predicts human preference between chatbot responses at 98% of state-of-the-art quality with 127x fewer parameters than the 9B+ competition-winning systems, which required 8x A100 GPUs and over \$100,000 of compute.
- Eliminated response position bias with swap augmentation and corrected roughly 20% overconfidence with post-hoc temperature scaling ($T = 1.20$), reaching validation log loss 0.9871 (6.2% better than baselines) and three-class accuracy of 52.06% (+3.5 percentage points).
- Trained the full pipeline on 57,477 pairwise human preferences in 8.4 hours on two free-tier Kaggle T4 GPUs at \$0 cost; ablations showed 30% of the training data already reaches 95.1% of peak performance.

ProbCLIP-A: Uncertainty-Aware Cross-Modal Retrieval via Probabilistic Adapters for Frozen CLIP Models -

Second author; contributed conceptualization, methodology, software and validation | Submitted to Elsevier, under review, 2026

- Designed a 4.2M-parameter probabilistic adapter (2.1M per modality) over a frozen CLIP backbone that learns Gaussian distributions over embeddings instead of point estimates, making retrieval confidence measurable.
- Achieved 68.9% text-to-image R@1 on Flickr30K with the best calibration of all baselines, cutting expected calibration error from 0.078 to 0.062 through Monte Carlo sampling; trained in 45 minutes on one free Kaggle P100.
- Uncertainty thresholding flags 69.3% of retrieval failures while touching only 4.9% of correct results, enabling practical human-in-the-loop review for high-stakes retrieval.

AWARDS AND COMPETITIONS

1st Place, Karnataka Education Datathon 2026 - Akshara Foundation | Aug 2026

Python, pandas, SciPy, statsmodels, Plotly, Streamlit

- Analysed item-level arithmetic assessment data for 1.38 million government primary school students across 31 Karnataka districts and 3 academic years, and ruled out infrastructure, gender and geography as primary drivers using rank correlation and variance decomposition.
- Isolated the driver at question level (matched pairs showed children failing an operation only once it required carrying or borrowing), built district Infrastructure and Education Access indices ranked by OLS regression (R-squared 0.56), and shipped a 7-screen Streamlit dashboard with a documented data quality report.

PROJECTS

Cachy - Short-Form Media Knowledge Engine

Python, FastAPI, Flutter, faster-whisper, Tesseract, SQLite, Hugging Face Spaces

- Built an end-to-end pipeline converting Reels, Shorts and articles into searchable knowledge cards using Whisper transcription, keyframe OCR and an LLM chain with automatic fallback across three providers (Gemini, Cerebras, Groq), served with a semantic knowledge graph from one free Hugging Face Space and an Android release.
- Diagnosed a production cost bug where a job queue polling once per second kept a serverless Postgres instance awake and exhausted the monthly compute quota in 17 days with zero users; replaced polling with event-driven wake-up and 30-minute backoff, reducing queries by 99.7% with no added latency.

IPL Match Outcome Predictor - Full MLOps Pipeline

XGBoost, FastAPI, Docker Compose, Streamlit, SQLite

- Deployed a three-service system on Docker Compose (FastAPI prediction API, drift monitor, live health dashboard) where the monitor computes Population Stability Index between training and live prediction distributions every 5 minutes and flags the model for retraining.
- Engineered 9 features (rolling win rates, venue history, toss decisions) and selected XGBoost over an MLP for equal accuracy with faster inference and interpretable feature importances.

Constitution of India - Retrieval-Augmented Question Answering

sentence-transformers, ChromaDB, Mistral-7B, Streamlit

- Wrote the full RAG loop from scratch in roughly 60 lines without LangChain (MiniLM embeddings, ChromaDB cosine top-5 retrieval, page-level citations), then stress-tested it with 20 adversarial queries, published the failure analysis, and raised answer accuracy from 65% to 78% by fixing chunk boundaries.

Bangalore Air Quality Hotspot Analysis

pandas, scikit-learn, DBSCAN, Jupyter

- Performed manual ETL on a year of non-standardized Excel exports from 14 CPCB stations (13 usable after quality checks) and engineered 6 domain features; of K-Means, hierarchical clustering and DBSCAN, only DBSCAN isolated critical hotspots as outliers instead of averaging them away (Silk Board, peak AQI 500).

Explainable Computer Vision - Plant Disease Detector and Multimodal Image Search

PyTorch, Grad-CAM, CLIP, Gradio, Hugging Face Spaces

- Trained a CNN over 15 plant disease classes with Grad-CAM overlays and severity grading so predictions are explainable, and built CLIP text-to-image and image-to-image retrieval with a per-query category breakdown chart; both deployed as public Gradio demos.

Additional shipped work (19 projects total, github.com/Vatsal057): AirSwipe, a MediaPipe webcam gesture controller for presentations with dwell timers and dead zones to suppress false triggers; TWIN, a real-time two-player game with no backend, secured by Firestore rules and transactional writes; ScribbleType, an on-device handwriting-recognition Android keyboard built as a Kotlin IME; and five native macOS apps in Swift.

EDUCATION AND CERTIFICATIONS

M.Tech, Data Science - M.S. Ramaiah University of Applied Sciences, Bengaluru, India | Nov 2025 - 2027 (expected)

B.Tech, Computer Engineering - SAL College of Engineering, Ahmedabad, India | 2021 - 2025 | CGPA 8.3 / 10.0

Certifications: IBM Data Science series on Coursera (5 courses, 2024).